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BYU CS 180 Lab 5: Cereal Data

Introduction:

Everyone loves cereal. But have you ever thought deeply about your cereal? Well now is your chance to take a data driven view of your breakfast.

Getting Started:

Download the data from github. Run the code below to download the data that you'll be using in this lab.

You may use pandas, numpy, matplotlib and/or seaborn for these excercises.

You can use/read their respective documentation in the links below (only if you need too, it's not required for the lab):

- Seaborn Documentation
- Matplotlib Documentation
- Numpy Documentation
- Pandas Documentation

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
cereal = pd.read_csv('https://raw.githubusercontent.com/porterjenkins/cs180-intro-da
```

EDA

```
cereal.head()
```

	name	mfr	type	calories	protein	fat	sodium	fiber	carbo	sugars	potass
0	100% Bran	N	C	70	4	1	130	10.0	5.0	6	280
1	100% Natural Bran	Q	C	120	3	5	15	2.0	8.0	8	135
2	All-	K	C	70	4	1	260	9.0	7.0	5	320

Next steps: Generate code with cereal | View recommended plots | New interactive sheet

notice that this list starts at 0

```
cereal.describe()
```

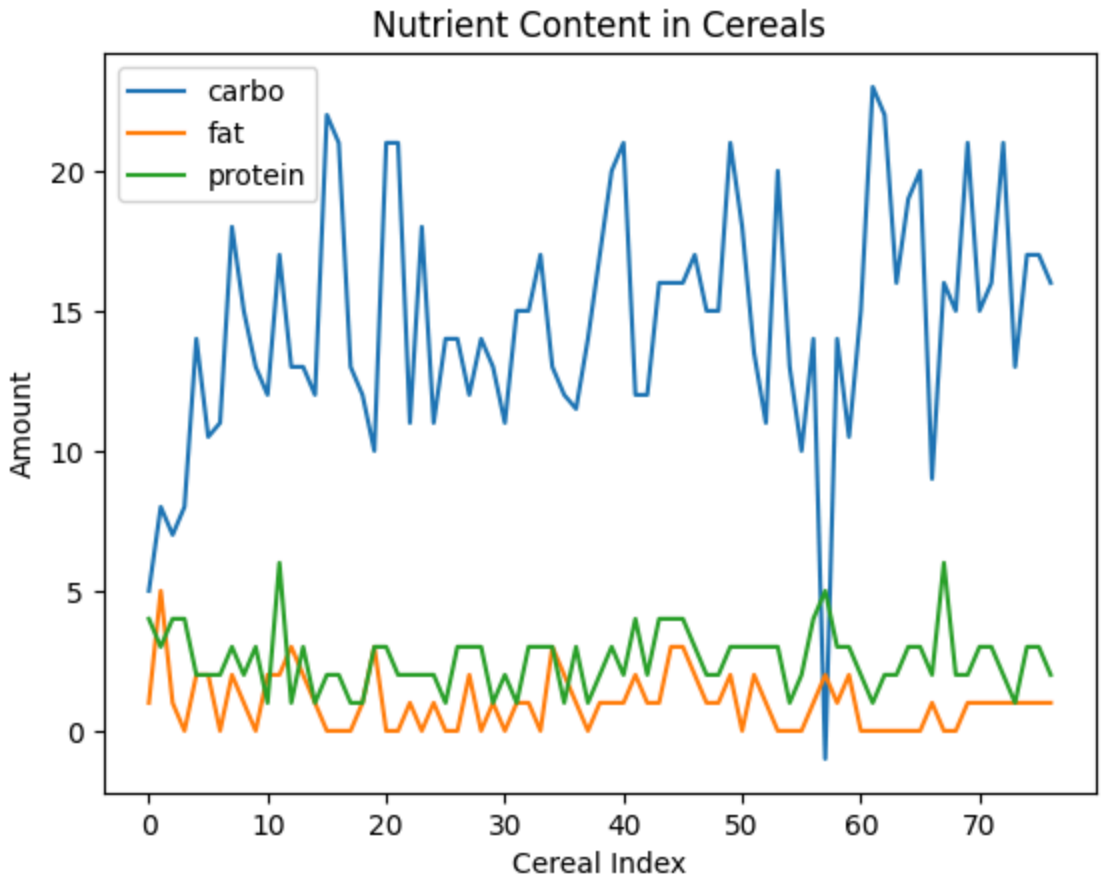
	calories	protein	fat	sodium	fiber	carbo	sugars	po
count	77.000000	77.000000	77.000000	77.000000	77.000000	77.000000	77.000000	77.0
mean	106.883117	2.545455	1.012987	159.675325	2.151948	14.597403	6.922078	96.07
std	19.484119	1.094790	1.006473	83.832295	2.383364	4.278956	4.444885	71.28
min	50.000000	1.000000	0.000000	0.000000	0.000000	-1.000000	-1.000000	-1.0
25%	100.000000	2.000000	0.000000	130.000000	1.000000	12.000000	3.000000	40.0
50%	110.000000	3.000000	1.000000	180.000000	2.000000	14.000000	7.000000	90.0
75%	110.000000	3.000000	2.000000	210.000000	3.000000	17.000000	11.000000	120.0

Exercise 1: Protein Powder

Carbs, fats and proteins are the three primary macro nutrients. Create a figure plotting the distribution of each of these macro nutrients together (i.e., three distributions on a single plot). Make sure to provide a legend.

```
# Enter all of your code for exercise 1 here. Feel free to add more cells if you nee
nutrients = cereal[["carbo","fat","protein"]]
for nutrient in nutrients.columns:
    plt.plot(nutrients[nutrient], label=nutrient)
plt.title("Nutrient Content in Cereals")
plt.xlabel("Cereal Index")
plt.ylabel("Amount")
plt.legend()
```

```
<matplotlib.legend.Legend at 0x7a5a93551ad0>
```



Exercise 2: Sugar Daddy

Get a list of the top 5 most sugary cereals and the 5 least sugary cereals.

```
# Enter all of your code for exercise 2 here. Feel free to add more cells if you nee
cereal.sort_values(by="sugars", ascending=False).head()
```

	name	mfr	type	calories	protein	fat	sodium	fiber	carbo	sugars	potass
66	Smacks	K	C	110	2	1	70	1.0	9.0	15	40
30	Golden Crisp	P	C	100	2	0	45	0.0	11.0	15	40
70	Total Raisin Bran	G	C	140	3	1	190	4.0	15.0	14	230

```
cereal.sort_values(by="sugars", ascending=True).head()
```

	name	mfr	type	calories	protein	fat	sodium	fiber	carbo	sugars	potas
57	Quaker Oatmeal	Q	H	100	5	2	0	2.7	-1.0	-1	11
55	Puffed Wheat	Q	C	50	2	0	0	1.0	10.0	0	5
20	Cream of Wheat	N	H	100	3	0	80	1.0	21.0	0	-

Exercise 3: Cereal Killer

Get a list of the top 5 highest rated and lowest rated cereals.

```
# Enter all of your code for exercise 3 here. Feel free to add more cells if you nee
cereal.sort_values(by="rating",ascending=True).head()
```

	name	mfr	type	calories	protein	fat	sodium	fiber	carbo	sugars	po
10	Cap'n'Crunch	Q	C	120	1	2	220	0.0	12.0	12	
12	Cinnamon Toast Crunch	G	C	120	1	3	210	0.0	13.0	9	
35	Honey Graham Ohs	Q	C	120	1	2	220	1.0	12.0	11	

```
cereal.sort_values(by="rating",ascending=False).head()
```

	name	mfr	type	calories	protein	fat	sodium	fiber	carbo	sugars	potas
3	All-Bran with Extra Fiber	K	C	50	4	0	140	14.0	8.0	0	33
64	Shredded Wheat 'n'Bran	N	C	90	3	0	0	4.0	19.0	0	14

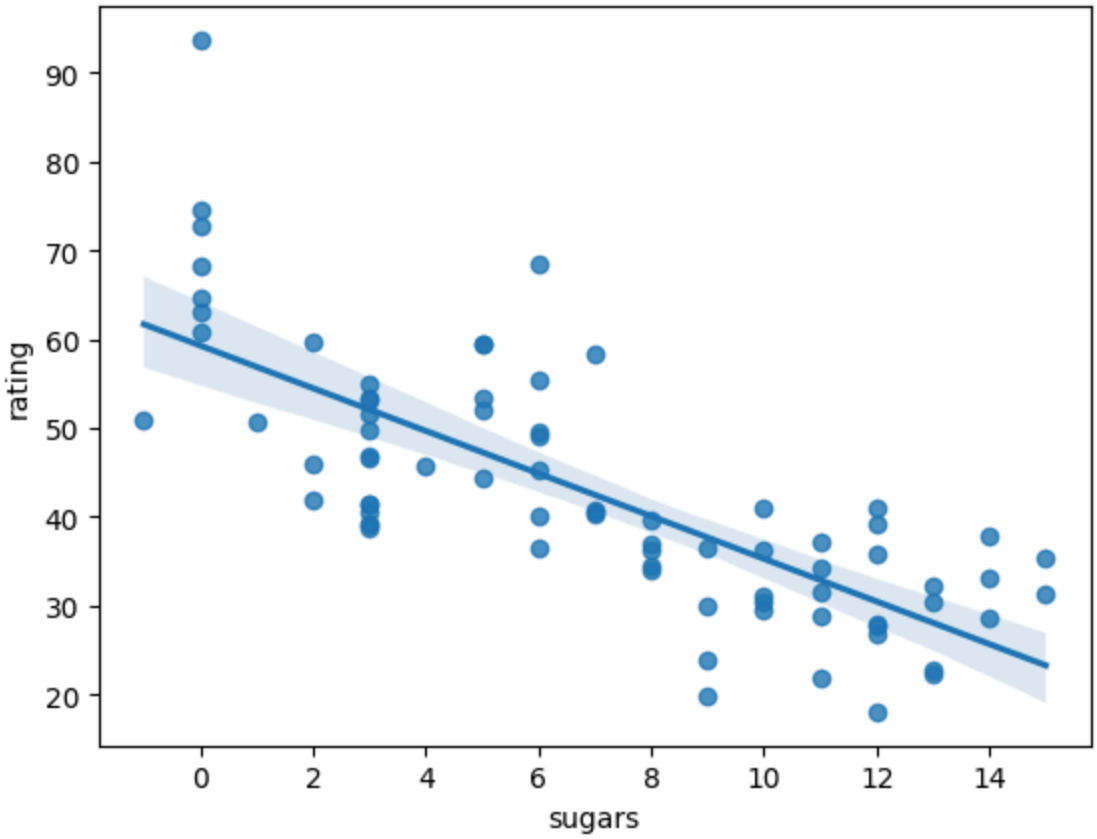
Exercise 4: America

Quantify the relationship between sugar and ratings.

Make a plot to visualize this relationship. Superimpose a best fit line (with seaborn) to describe the relationship. It may be helpful to look at the seaborn regplot documentation.

```
# Make the plot for the data visualization and line of best fit here:
sns.regplot(data=cereal,x="sugars",y="rating")
```

```
<Axes: xlabel='sugars', ylabel='rating'>
```



Calculate a correlation statistic describing the relationship between sugar and ratings (i.e., r or r squared).

```
# Calculate the statistic using this cell:
cereal["sugars"].corr(cereal["rating"])
```

```
-0.7596746584301076
```

Write a statement in plain English interpreting this statistic.

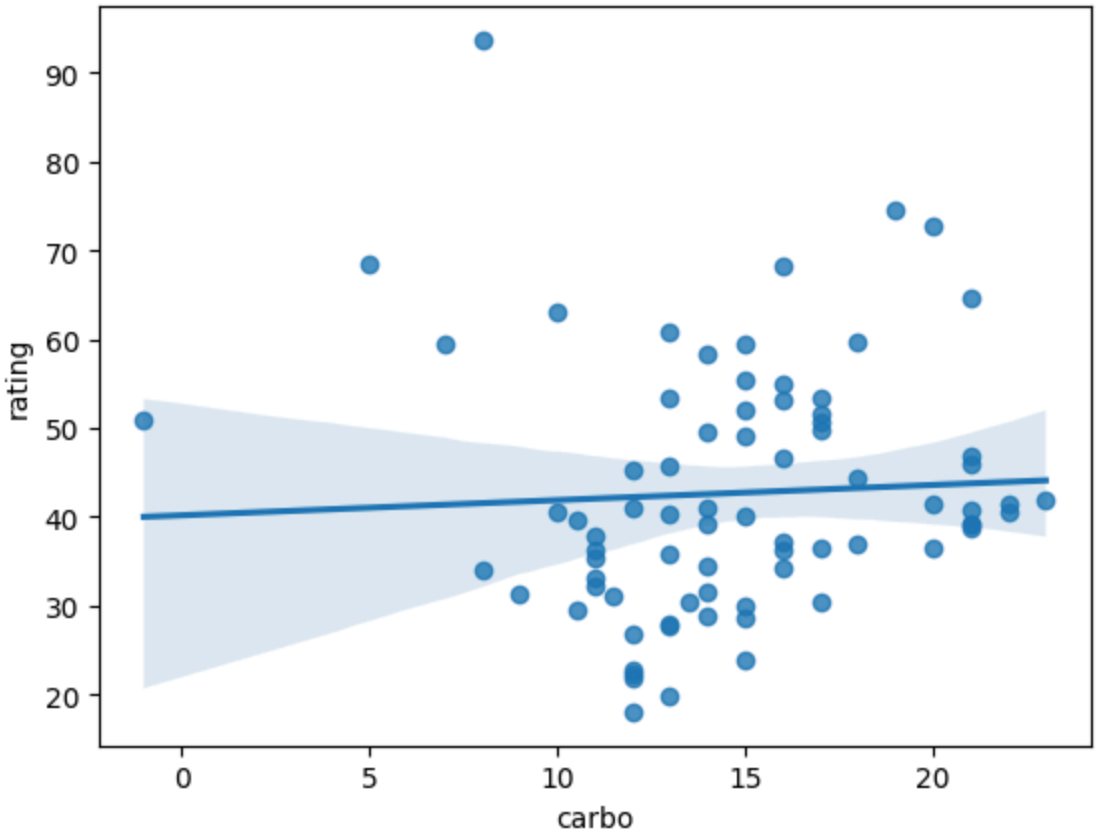
(Write your statement here) There is a poor correlation between the sugar content and the rating that a cereal will receive. It would be best to find a different variable to explain ratings. This correlation can be interpreted as the more sugary cereals tend to have receive a lower rating. Because it is such a significant negative correlation, this shows an inverse relationship.

Exercise 5: America Part 2

Make five plots comparing the relationships of carbo, sugars, calories, protein, and fat with rating.

```
# Write your code to compare the various variables with rating below:
sns.regplot(data=cereal,x="carbo",y="rating")
```

```
<Axes: xlabel='carbo', ylabel='rating'>
```

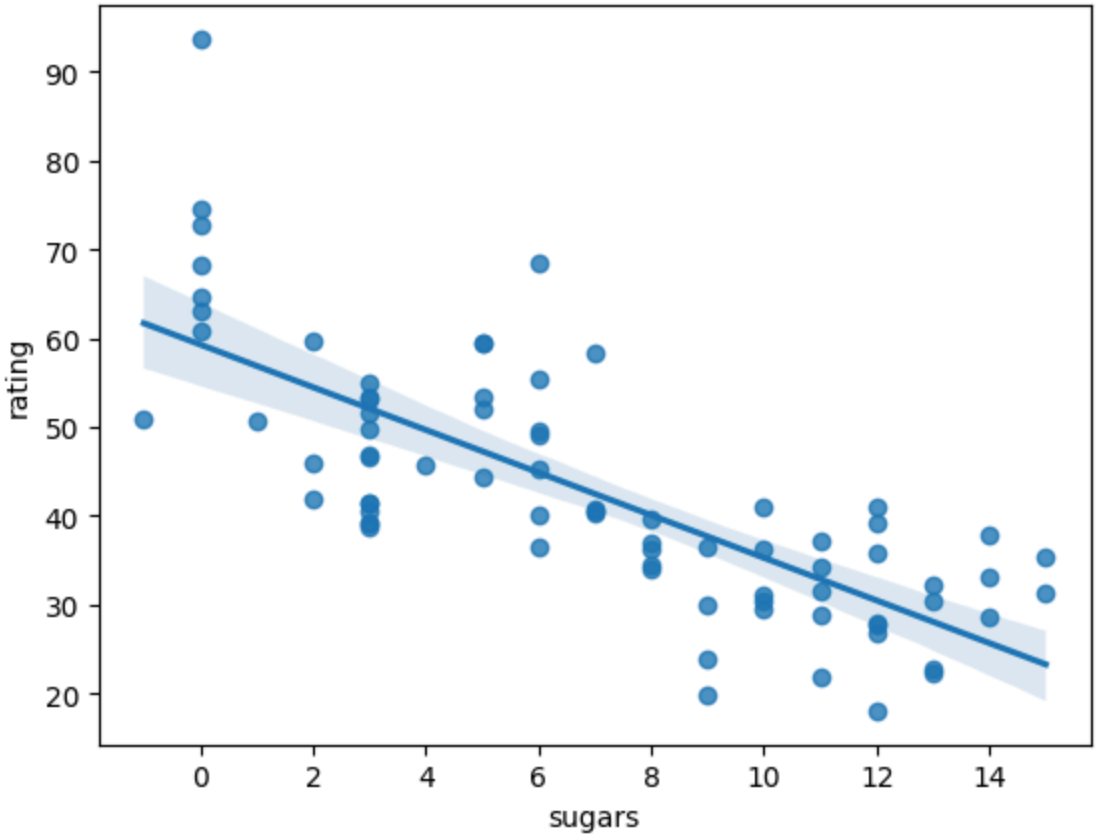



```
cereal["carbo"].corr(cereal["rating"])
```

```
0.05205466123376461
```

```
sns.regplot(data=cereal,x="sugars",y="rating")
```

```
<Axes: xlabel='sugars', ylabel='rating'>
```

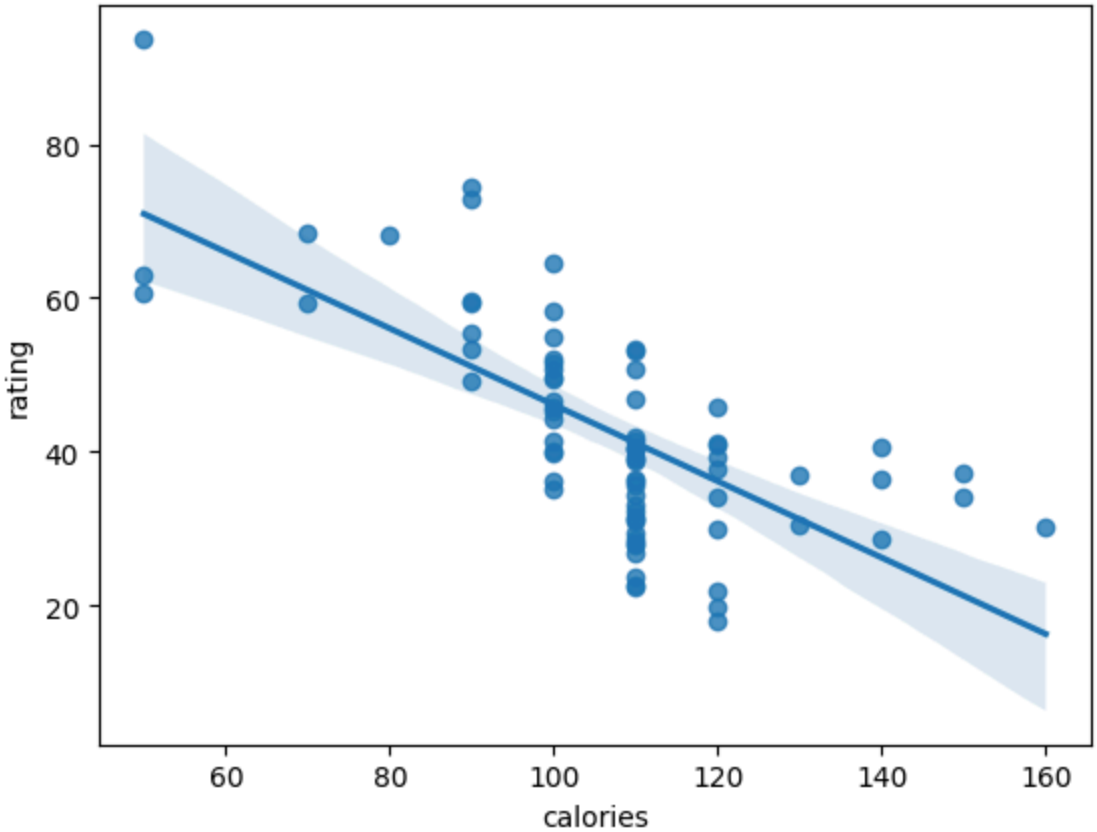



```
cereal["sugars"].corr(cereal["rating"])
```

```
-0.7596746584301076
```

```
sns.regplot(data=cereal,x="calories",y="rating")
```

```
<Axes: xlabel='calories', ylabel='rating'>
```

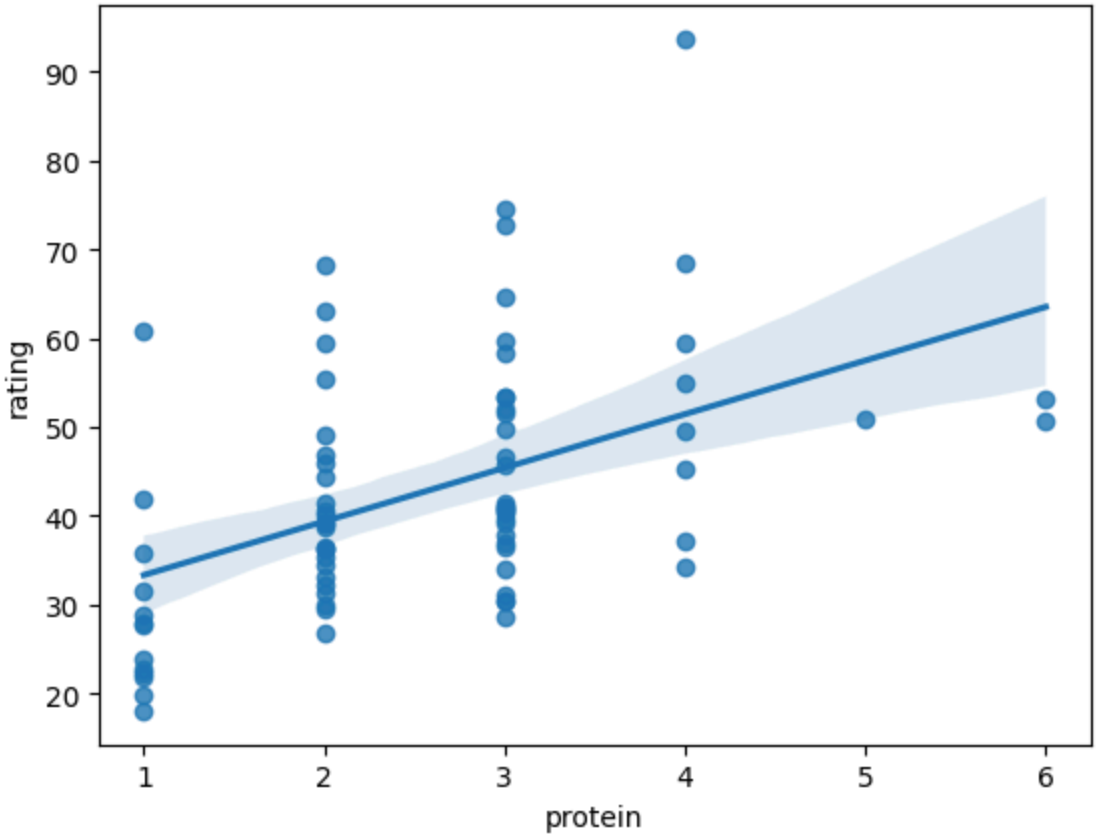



```
cereal["calories"].corr(cereal["rating"])
```

```
-0.6893760311652586
```

```
sns.regplot(data=cereal,x="protein",y="rating")
```

```
<Axes: xlabel='protein', ylabel='rating'>
```

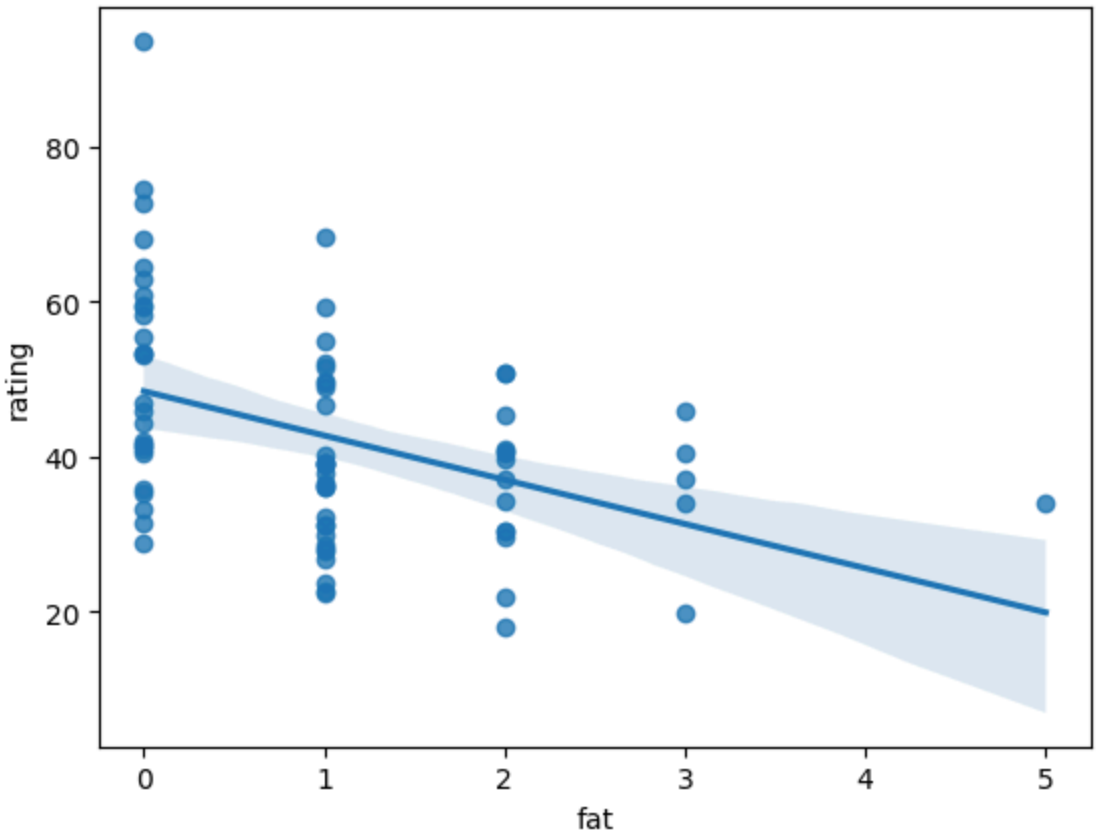



```
cereal["protein"].corr(cereal["rating"])
```

```
0.4706184645492595
```

```
sns.regplot(data=cereal,x="fat",y="rating")
```

```
<Axes: xlabel='fat', ylabel='rating'>
```




```
cereal["fat"].corr(cereal["rating"])
```

```
-0.40928366014781
```

Of the variables carbo, sugars, calories, protein, and fat, which has the strongest relationship with rating? Justify your answer.

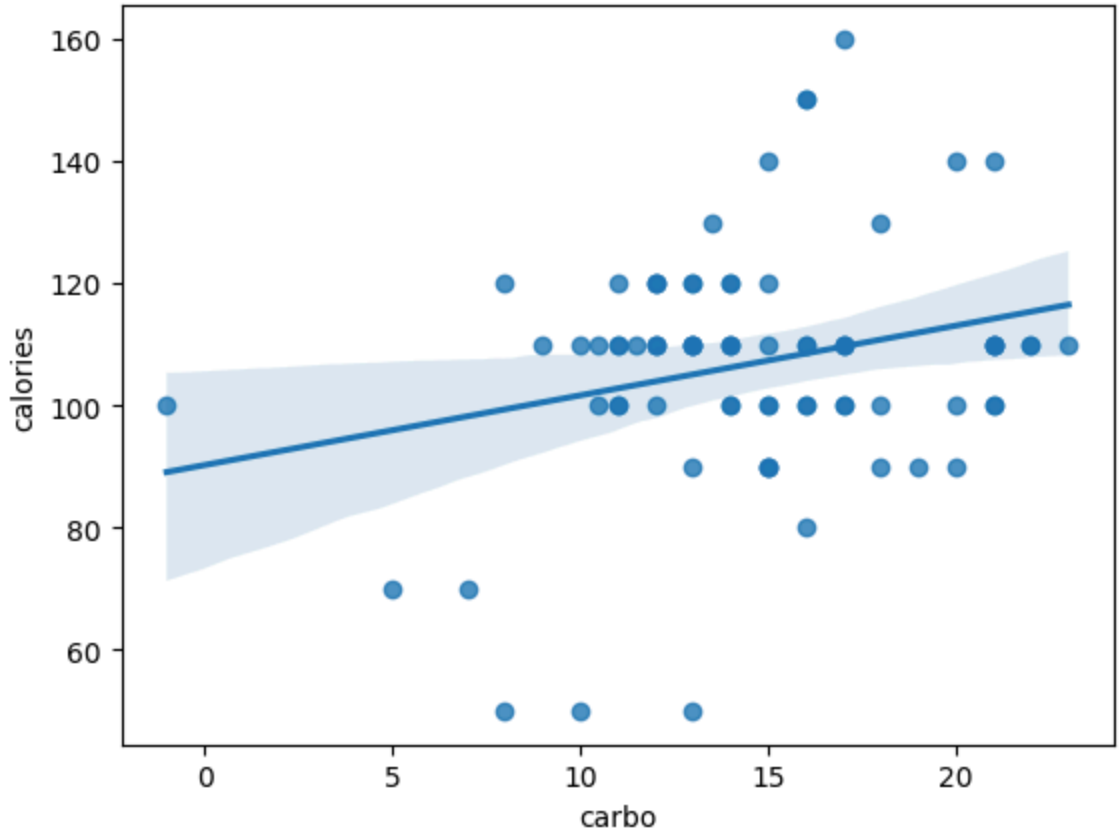
(Write your statement here) Protein has the highest correlation as it is the only positive correlation.

Exercise 6: Preparing for Mt. Everest

Do the same as you did with exercise 5, but instead compare carbo, sugars, protein, and fat with calories.

```
# Write your code to compare the various variables with calories below:
sns.regplot(data=cereal,x="carbo",y="calories")
```

```
<Axes: xlabel='carbo', ylabel='calories'>
```

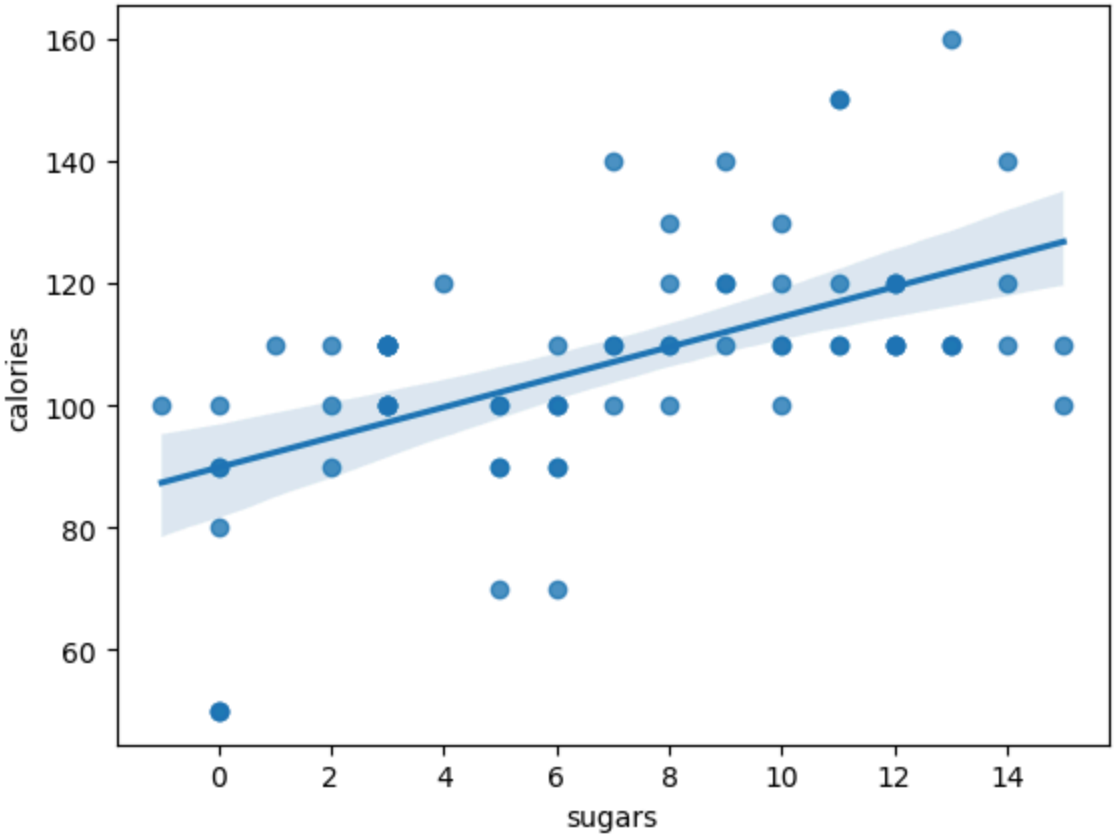



```
cereal["carbo"].corr(cereal["calories"])
```

```
0.25068091469892756
```

```
sns.regplot(data=cereal,x="sugars",y="calories")
```

```
<Axes: xlabel='sugars', ylabel='calories'>
```




```
cereal["sugars"].corr(cereal["calories"])
```

```
0.5623402898034883
```

```
sns.regplot(data=cereal,x="protein",y="calories")
```

```
<Axes: xlabel='protein', ylabel='calories'>
```

```
cereal["protein"].corr(cereal["calories"])
```

```
0.019066068232357035
```